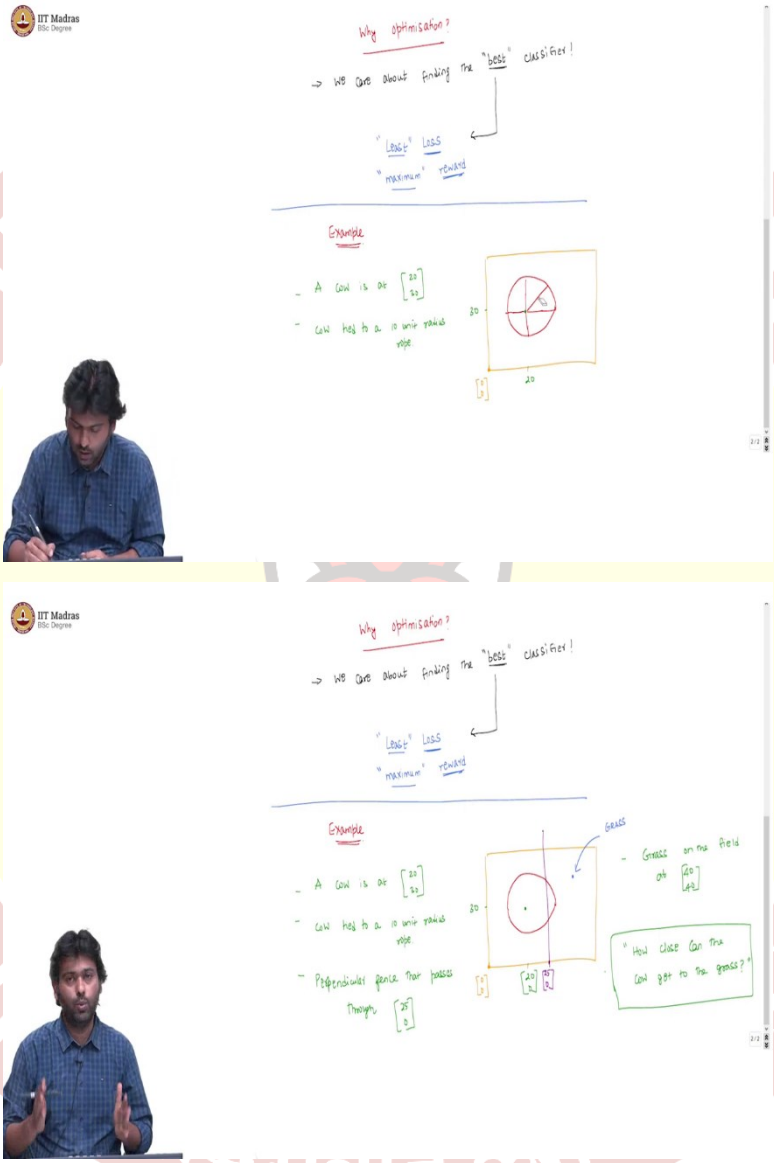


IIT Madras

ONLINE DEGREE

Machine Learning Foundations
Professor Arun Rajkumar
Department of Computer Science & Engineering,
Indian Institute of Technology Madras
Introduction to Optimization

(Refer Slide Time: 00:13)



Welcome back. So, we are trying to understand why optimization. And we said that the reason why we need optimization is to convert data to decisions. So, in machine learning why do we care about optimization? Well, one way to answer this would be to say that we always care about finding, let us say, for example, if we are in supervised learning, we care about finding the best classifier.

So, there are different ways to classify, let us say, an email a spam or not spam. And given a data where some data points are spam emails and some data points or non-spam emails and now given a new email, you want to understand whether this email is spam or not and there are so many classifiers that can take an input email and then classify it as spam or not spam, but somehow we need to understand which is the best classifier. So, that is this notion of best that comes in. But then best is kind of a vague notion. I mean, just describing it like that. So, we need to understand what do we really mean by best?

So, often, when I say best, it means that, well, there are multiple ways of describing best depending on what you want to do. One particular example could be the best classifier is the one that has some kind of least loss. Well, you define something called as a loss, and then among all possible classifiers, you look at the classifier that has the least loss, and the one that has the least loss is what is called as the best classifier. Of course, now, that begs the question, what do you mean by a loss?

So, well, the specific way of defining loss will depend on what problem you are looking at, what specific paradigm in machine learning you are looking at and so on. And that is not the goal here. But I wanted to give a brief view that, well, in machine learning, you are always after the best something, it could be a classifier, best representation for data, whatever it is. And in some sense the best would translate to either the least something, could be the least loss, or in some cases, it could be the maximum reward. And again, I am not defining what reward is here. It depends on the problem at hand. And we will see a lot of examples as to what this loss could be and what these rewards could be as we go along in this course.

For the moment, let me leave it at that and say that optimization is extremely fundamental to understand machine learning algorithms, because machine learning algorithms always want to make a decision in the best possible way and best is either minimizing loss or maximizing reward and things like that. So, there is always least or maximum that comes into picture.

So, with that introduction to why optimization is relevant to machine learning, now we are going to take a step back and understand the basics of optimization itself. So, what does it mean to say you are optimizing something and how do you optimize it? So, what is the algorithmic way to go about optimizing whatever we want to optimize?

So, to do this, we are not going to think about machine learning at the point. At this point, we are going to think about a more broad framework of optimization and we will try to understand optimization in a more general way. Of course, we will relate it to specific machine learning

problems as we go along in this course, but for the moment, we will go to step back a bit and say that, hey, I am going to understand optimization itself as a general mathematical approach and then we will try to apply it to machine learning problems.

So, with that, let us start with a simple example. And I hope this example will illustrate the different things that one needs to keep in mind while coming up with an optimization problem. So, the example is a cooked up example. Nevertheless, it illustrates what we want to do very well. So, let us try to understand what this example is. So, let us say you have a field, so an acre of field. Let us say, this is a square field, which we are going to think of this as a 2 cross 2 grid, where the left bottom part is what is the origin. So, this is the point 0 , 0.

Now, what happens in this field? Well, at some point in this field, let us say, the point 20 , 30, there is a cow. A cow is at 20, 30. So, this is the point which corresponds to 20 on the x-axis, 30 on the y-axis and that is the position of a cow. Now, what is this cow doing? So, well, the cow is tied to, let us say, a one, let us say, a rope. So, cow tied to a, let us say, 10 unit radius rope. So, there is a rope that you have tied to the neck of the cow and the length of the rope is, let us say, 10 units, in whatever units we are measuring this fields length itself.

What does that mean in this picture? Well, if the cow is at 20 , 30 and if it is tied to a rope at, which is of length 10 units, then where are the positions where the cow can reach in this field, on this field? So, pause for a second and try to think about that problem. It is not hard. But I would encourage you to pause and think.

Well, because the rope is 10 units long, the cow is going to is, going to be able to move 10 units in any possible directions in the field. And it cannot move more than 10 units. So, let me make this slightly larger to make the point clearer. So, now if you want to understand the set of all points where the cow can move to, well, it is going to simply be a circle. My circles are bad, but hope that should be okay. Anyway, imagine that a circle and of radius 10. So, that is where the cow is allowed to move, simply because it is tied to this rope, which is 10 units long. I erase this part and just leave the circle there.

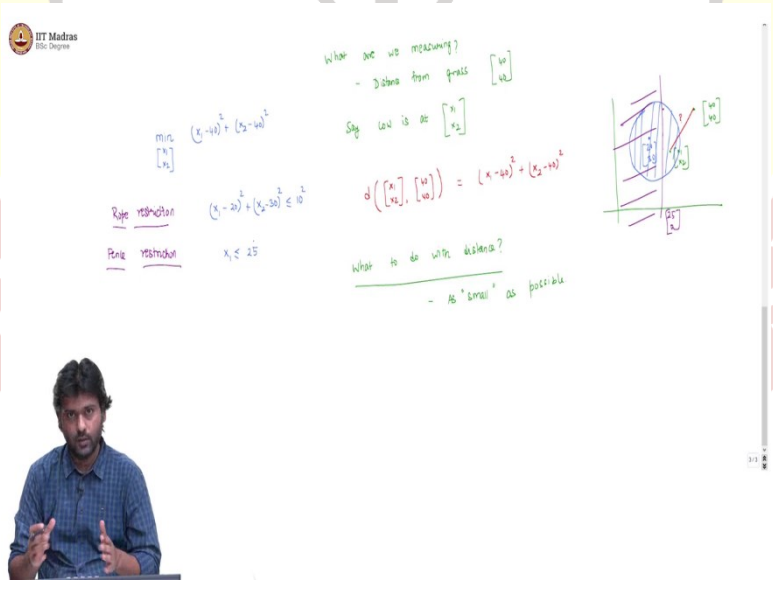
Now, what else? What are, what else is happening in this picture? Well, the other thing that is there in this picture is that there is a perpendicular fence. So, there is a perpendicular fence that passes through, let us say, the point 30 or rather, let me say, let us say, the point 25, 0. So, there is a perpendicular fence. Now, let us try to understand that in this picture using a different color. So, there is a perpendicular fence, let us say at point 25 , 0, so this is 25, 0. And of course, the previous point is 20, 0.

Now, how does the perpendicular fence look like? Well, it is going to look like this. It is going to be a straight line. And because of this fence, the cow will not be able to move further away from this fence. So, that is another restriction so to say on the cow. So, now what is the, where are we getting at with all this? Now, I am also going to say that there is one more point here. So, there is some grass on the field at, let us say the position 40, 40. So, where is 40, 40? Well, 40, 40 is somewhere here and this is grass. This is the position where you have grass.

So, this is the setup. You have a cow and the cow is tied to a rope and there is a perpendicular fence running at some point in the field and there is grass at some point in the field. Now, in this setup, the question that I am going to ask is how close can the cow get to the grass? So, this is the question that I want to answer. So which means that the cow is allowed to move wherever it is allowed to move with its restrictions with respect to the rope and the fence.

With these two restrictions, it can reach to some point where it is closest to the grass. Maybe it reaches the grass itself. Well, in that case, it will eat the grass. Otherwise, it will reach as close to the grass as possible. So, now the question we are asking is, how close can the cow get to the grass? And we want to understand this in a more mathematical framework and that is what we are going to do next.

(Refer Slide Time: 10:48)



So, we have set up the problem. So, now we want to understand what exactly are we measuring? So, what are we measuring? What are we, what is our goal? So, what are we trying to find? Well, the question asked was, how close can the cow get to the grass or the fodder? Which means that we are measuring distances in some sense and what is exactly the distance that we

are measuring? Well, we want to measure the distance from the grass. So, the distance from grass, that is what we are trying to measure. And of course, we said that the grass is at 40, 40.

Now, for any position x , y of the cow, or let me call it x1, x2 of the cow, we are trying to measure how close it is, is that position to the position 40 , 40. So, say cow is at the position x1, x2, where x1 is the horizontal axis and x2 is the vertical axis as measured from the origin that we looked at earlier. Now, if the cow is at x1, x2 and we wanted to understand its distance from the grass which is at 40, 40, well, how do we find this distance? Well, the 40, 40 point is somewhere here, 40, 40 and let us say the cove is at some x1, x2 and we want to understand what is this length. So, how much is this?

Well, this is easy to compute. So, this is simply the distance of x1, x2 , 40, 40 and I am sure all of us has seen this at some point in our schools. So, this is simply x1 minus 40 squared plus x2 minus 40 squared, of course, that is a square root there which will be the distance, but let us think of this as the squared distance, which means that it is simply x1 minus 40 squared plus x2 minus 40 squared. So, now what we want to do with this distance. So, the next question is, what do we want to do? So, what to do with this distance?

Well, our goal is to make this distance as small as possible. So, that is our goal, because we want the cow to get as close to the grass as possible, which means that we want the distance to the grass to be as small as possible, which means that we want to find that position of cow which will have the least distance to the grass. Well, if you think of the position of cow as x1, x2 then the distance of the grass is exactly what we just gave. So, which means that our goal then, let me read it here, is to minimize over x1 and x2, x1 minus 40 squared plus x2 minus 40 squared.

Well, is that all it? So, if I had to find the answer to this problem, it means that I am asking, well, the cow is at some point, but then how close can it get to the grass. I have not taken into account the rope yet and they have not taken into account the fence set. If I do not take that into account, of course, the cow does not have a rope and there is no fence when of course the cow can reach the grass itself, which means that the answer to this would be the minimum distance is 0 and the point is 40 , 40. So, but then we do have some restrictions on the movement of the cow. And we have to somehow incorporate those restrictions into the problem.

Well, there are two restrictions. The first restriction is what we call as the, well, the rope restriction. The second one is the fence restriction. Let us try to understand what is the rope restriction do to our problem. Well, the rope restriction would tell me that I cannot choose

arbitrary x1 and x2 as positions of the cow, because the rope is restricting me only certain x1 and x2 where the cow can actually be present. And these restrictions we know that if the cow is at 20, 30, then the rope restriction would only allow me to move to a radius of 10 units. So, which means that, how do I mathematically formulize that?

I formalize that by saying that whatever x1, x2 that I finally find as answer, it necessarily has to satisfy x1 minus 20 squared plus x2 minus 30 square. Well, what am I trying to say? I am trying to see what is the distance from 20 , 30, which is the initial position of the cow to where all it can eventually end up with.

So, we know that it has to be in this region of the circle, which means that this value necessarily has to be less than or equal to the square of the radius. So, in this case the radius is 10 units long, so which means that this is going to be less than or equal to 10 square. So, this is what the rope restriction is forcing me to choose an x1 and x2 with certain constraints. So, this is the rope restriction.

Of course, there is one more restriction which is the fence restriction. Now, what is the fence restriction say? The fence restriction says that there is a perpendicular fence that runs at the point so 30 ,, sorry, that runs through the point 25 , 0. So, which means that, if I have to convert that into an equation, now what does the fence restriction tell me in terms of x1 and x2, the set of all positions where the cow can could, can potentially could be.

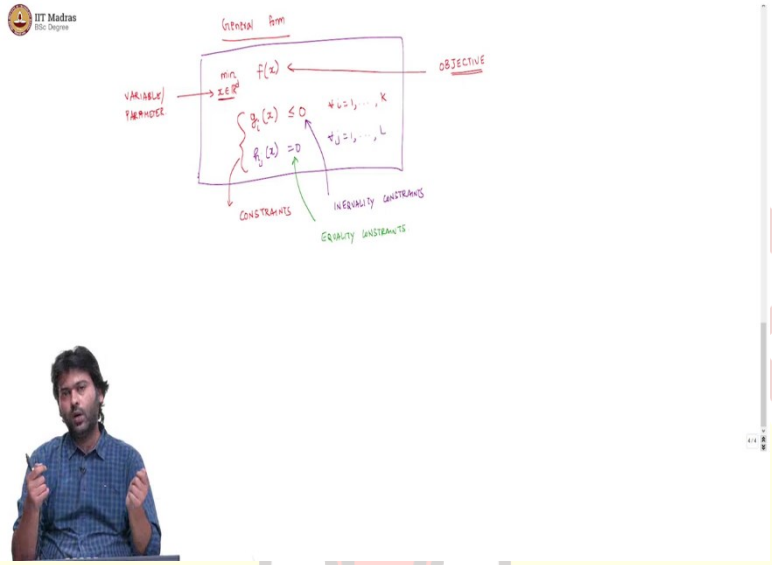
Forgetting the rope for the moment, if you only focus on the fence, well, the fence is passing through 25 , 0. Well, it can be, the cow can be anywhere on this region, in this region. So, that is what the fence restriction says. It cannot pass the fence, which means that it is okay to be anywhere on the left side of the fence.

Now, how do I codify that in terms of equation? Well, I would simply say that the fence restriction says, well, the only restriction that it is putting me is on the horizontal axis. I do not care where the cow moves in the vertical axis, at least the fence does not restrict the cow to move, cow's movement in the vertical direction. It only restricts it in the horizontal direction, which means that the restriction is that x1, which is the horizontal direction in which the cow can move, has to be necessarily less than or equal to 25.

So, now, we have converted our problem into a mathematical problem which can be solved. So, we have the problem as follows minimum x1, x2, x1 minus 40 squared plus x2 minus 40 squared such that two things happen x1 minus 20 square plus x2 minus 30 square is less than

100, less than or equal to 100 and x1 is less than or equal to 25. So, now, this is an example of an optimization problem. Now, we have set up the optimization problem. Now, we will understand it in a more general framework next.

(Refer Slide Time: 18:42)



So, now we are going to understand whatever optimization problem we just set up in a more general form. So, the general form of an optimization problem then is as follows. We want to minimize some function of x. So, this x can be, in general, let us assume this x can be anywhere in RD, the D dimensional real space. In the cow example, this was a two dimensional space. In our example, it can be anything. So, in the, sorry, in the general form it can be anything, x can be RD.

Now, we also had some restrictions on the moment of cow now that will translate in the general form as, well, there are some constraints on what x could be. And one way to say this is to say that there are a bunch of constraints functions which determine where x can lie. So, on these constraints, in this case, our g1 of x is less than or equal to 0, g2 of x is less than or equal to 0 and so on till gk of x less than or equal to 0.

So, in the previous example, in the cow example there was only one, there were only two constraints, one was the rope constraint, other was the fence constraint, which means that g1 of x would be the rope constraint and let us say g2 of x would be the fence constraint.

In addition to this, there may be problems where you also have a constraint which is not a less than or equal to constraint, but it might be a equal to 0 constraint. So, the constraint could be hj of x equal to 0 for all j equals 1 to L. Think of it as maybe you have a specific value for a

budget and you want to spend that much amount and that can be, for instance, codified as an equality constraint. So, this is a very general form of an optimization problem.

Now, it has a lot of components. Let us just give these components names so that in our interaction, as we go along, we will use these names so that it becomes clear what we are talking about. The function that we are trying to minimize in this case is what we will call as the objective. So, what is our objective in the previous case, in the cow example, our objective was to minimize the distance to the grass, so that was our objective.

Now, the value that you will eventually end up finding, so your parameter of interest is what we will call sometimes as variable on which you are optimizing or sometimes it is also in machine learning parlance, this will also be called as parameter.

So, in the case of the cow example, again, this is the position of the cow, so, and that was a two dimensional parameter. In general, it could be a D dimensional parameter. Now, what are these? Well, these restrict the value that the parameters can take and we are going to call these as constraints. Specifically, we will call the one, the constraints where j of x is less than or equal to 0 as inequality constraints and the case where j of x is equal to 0 as equality constraints.

Now, here is a quick question for all of you. Well, the way I have written this has a certain form. So, I am trying to minimize an objective subject to certain inequality and equality constraints. And the inequality constraints are always of the form j of x is less than or equal to 0. And the equality constraints are always of the form hj of x equal to 0. Now, in the cow example, there was no 0 on the right hand side. So, we said that the inequality constraint was that the distance, the cow is tied to a rope, which is of radius 10, and we converted it to an equation.

Now, this is what I am claiming as a general form what you are seeing now. Now, the question to you is, can you always put any inequality constraint into this form and any equality constraint into this form? That is a simple question. But then think about it. The second question is that this, the form that we have stated here is an objective which we want to minimize.

In some cases, if the objective is reward, you might want to maximize it. So, now, do we need a separate whole theory for that or do you think is it possible to convert a maximization problem into an equivalent minimization problem? If so, how would you do about it? Go about doing it. So, think about that as well. So, these are two questions for you to ponder and think about.

So, with that, we have given the general form of optimization problem. Now, it begs the question, well, fine, now, we know how to convert our problem into an optimization problem, we understand the basic structure of an optimization problem, now how do we solve the problem? So, that is the most important part.

Well, I know how to set up the cow's problem as a proper mathematical question, optimization problem in the general form, but how do I go about solving this. So, how do I finally get the answer? What is the algorithm to get the answer? And that is what we are going to start developing next.

